

Exp. No: 3 Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

Aim:

To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

Software required:•SCILAB version 6.0.1

Practical Calculation:

```
1 clc;
2 clear;
3 clf();
4 charge_1 = 3*10^-6;
5 charge_2 = 2*10^-6;
6 radius = 3*10^-2;
7 current_1 = 3;
8 current_2 = 2;
9 l = 4*10^-2;
10 d = 1*10^-3;
11 mu = 4*pi*10^-7;
12 epsilon = 8.854*10^-12;
13 F = (charge_1*charge_2)/(4*pi*epsilon*radius^2);
14 f = (mu*current_1*current_2*l)/(2*pi*d);
15 disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES");
16 disp(F);
17 disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS");
18 disp(f);
19 x = linspace(0,10,50);
20 y = F*exp(-0.1*x);
21 z = f*exp(-0.1*x);
22 subplot(1,2,1);
23 plot(x,y);
24 xlabel("R");
25 ylabel("Force");
26 title("FORCE BETWEEN TWO ELECTRIC CHARGES");
27 subplot(1,2,2);
28 plot(x,z);
29 xlabel("D");
30 ylabel("Force");
31 title("FORCE BETWEEN TWO PARALLEL CONDUCTORS");
```

Scilab 6.1.1 Console

"THE FORCE BETWEEN TWO ELECTRIC CHARGES"

59.918283

"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"

0.000048

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Theoretical Calculations:

③ Force of attraction b/w two electric charges and two parallel conductors.

1) Force b/w two electric charge $F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$

2) Force b/w two parallel conductors $f = \frac{\mu_0 I_1 I_2 l}{2\pi d}$

$q_1 = 3 \times 10^{-6}$
 $q_2 = 2 \times 10^{-6}$
 $r = 3 \text{ cm}$
 $I_1 = 3 \text{ A}$
 $I_2 = 2 \text{ A}$
 $l = 4 \text{ cm}$
 $d = 1 \text{ mm}$

1) Force b/w two electric charges $F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$

$$F = \frac{(3 \times 10^{-6})(2 \times 10^{-6})}{4 \times 3.14 \times 8.854 \times 10^{-12} \times (3 \times 10^{-2})^2}$$

$$= \frac{6 \times 10^{-12}}{1.001 \times 10^{-13}}$$

$$F = 59.92 \text{ N}$$

2) Force b/w two parallel conductors.

$$f = \frac{\mu_0 I_1 I_2 l}{2\pi d}$$

$$f = \frac{(4 \times 3.14 \times 10^{-7})(3)(2)(4 \times 10^{-2})}{2 \times 3.14 \times 1 \times 10^{-3}}$$

$$= \frac{3.0144 \times 10^{-7}}{6.28 \times 10^{-3}}$$

$$f = 4.8 \times 10^{-5} \text{ N} \Rightarrow 0.000048$$

Result: Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.